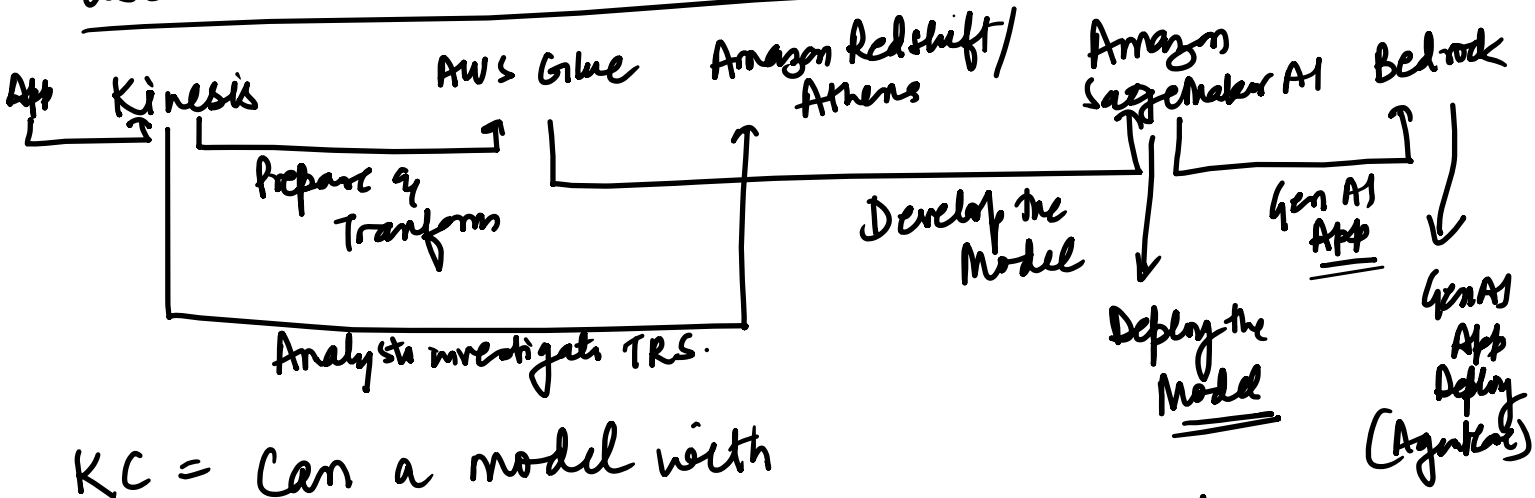
Day 1 = Setup & Data Preparation using SageMaker AI
Lab 1 to 5

Day 2 = Model TRG & Deployment
Lab 6 to 10

Day 3 = Monitoring, Updates, Capstone Project
5 to 6 HRs.

Use case: Fraud detection



KC = Can a model with excellent prediction accuracy still create business problem?

Yes / No — Yes = suffer data drift, lack transparency, unfair behavior, overfitting

Reasons for NB failure: Local Machine = NB.

1. Python versions
2. Missing packages
3. incompatible dependencies
4. insufficient memory

④ Insufficient memory
⑤ differences in OS.

SageMaker AI = ML lifecycle
SageMaker Studio = Dev env
↳ Domain
↳ user profile
↳ Space (JSpace)
↳ Notebook
↳ user profile

Module 2:

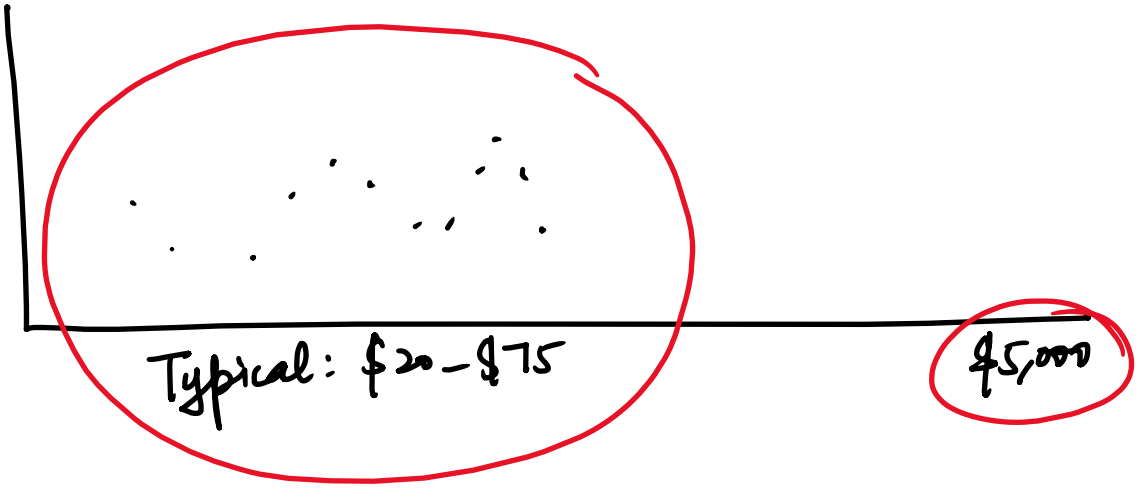
[AnyCompany Consulting = Enterprise Customer Intelligent Platform
ORG = [Customer records, TRANSACTIONS, prod info, Historical Interaction data → S3, DB, env.]
↳ Data Scientists — 5 challenges
① Incomplete, Inconsistent & duplicated data
② Large Datasets — Distributed processing
③ Existing Transformation scripts — in AWS
④ Repeated feature engineering across DS Teams
⑤ Need for consistent features during both Model

(5) Need for consistent features during both ...
TRGs & Rod Inference.

"Outliners, Class Imbalance, Target Leakage"

(1) Outliners

↳ Data point that differs significantly from the majority of observation in a dataset.



Why outliners matter?

(1) They can distort the mean, SD, model TRGs.
(2) Represent rare but imp Business events – fraud / equipment failure

How to handle outliners?

You can remove records confirmed to be erroneous, apply a log transformation.

(2) Class Imbalance

↳ when one target class has significantly more observations than another.

bank – 10K transaction

observations

Suppose a bank — 10k transaction

[Model Predict = 99% accuracy]

③ "Target leakage" (Data leakage)

↳ when a model uses information that would not legitimately be available when making a prediction, or information that directly / indirectly reveals the target

Classification Eval Metrics

↳ ML model predicts categories / classes.

↳ 2 class { Positive (1) ⇒ fraud Trans
Negative (0) ⇒ legitimate Trans.

Accuracy, Precision, Recall, F1-score, ROC, AUC

1. Confusion Matrix = Compares actual classification with Model predictions

1000 Transactions = 100 are fraud
900 = Not fraud

Confusion Matrix — fraud Detection

Actual ↓ / Predicted →	Fraud (1)	Legitimate (0)
Fraud (1)	✓ True Positive (TP) (Correctly detected) 80 ✓	False Negative (FN) (Missed fraud) 20
Legitimate (0)	False Positive (FP) (False alarm) 50 ✓	✓ True Negative (TN) (Correctly cleared) 850 ✓

50 ✓ 850 ✓

(2) Accuracy = How often is the model correct ?

$$= \frac{TP + TN}{TP + TN + FP + FN} = \frac{80 + 850}{1000} = 93$$

930/1000

↳ Misleading when classes are misbalanced.

(3) Precision = When the model predicts fraud, How often is it correct

↳ Reliability of Positive predictions

$$= \frac{TP}{TP + FP} = \frac{80}{80 + 50} = 61.54\%$$

(4) Recall = How many actual fraud cases did we catch?

↳ Sensitivity / True Positive Rate (TPR)

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{80}{80 + 20} = 80\%.$$

(5) F1 - Score

↳ $$F1 = \left[2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right]$$

OR

$$F1 = \frac{2TP}{2TP + FP + FN}$$

$$= \frac{2(80)}{160 + 50 + 20}$$

$= \frac{\quad}{160 + 50 + 20}$

Model	Precision	Recall	F1-Score
A ✓	95%	40%	56.3%
B ✓	70%	75%	72.4%
C ✓	60%	95%	73.5%

How Apache Spark works.

SPARK DRIVER
(Plans & Coordinates distributed work)

Worker 1 ✓ | Worker 2 ✓ | Worker 3 ✓
Tasks/partition | T/P | T/P

Processed output

Input data ✓ (Amazon S3) → Processing Job (Script + Container) → output ✓ (S3)

[Job status & logs.]

Day 1 = Setup of Sagemaker AI (Service Catalog)

Day 2

[Data Prep] → ML Model Training ✓ → ML Deployment → Monitoring

Prep ↓ Training

① Canvas ✓ (No-Code / Low-Code) — Data Wrangler
② EMR ✓ → Cluster
③ SageMaker Processing (70-20-10) (80-10-10) (70-25-5 Tr Te Val)

Mod 3: Model Development

① Track the Iteration → Lab 6
② Tune Models using SageMaker Experiment
③ Analyze, Detect, Set Alerts → Lab 7 — SageMaker Debugger
④ Bias in ML Models → Lab 8

Train ✓ Test ✓ Validation ✓

④ features (Columns)
↳ feature Stores → Lab 5

Which processing tool would you choose ?

① Transform millions of dates to days of the week rapidly
↳ EMR with spark → large distributed workload.

② Identify seasonal variation in water level readings
↳ Data Wrangler → Time series

③ Combine multiple fields into a custom binary "feature" (Column)
✓ Data Wrangler → Custom Transformation
✓ SageMaker Processing → Custom Script

Amazon SageMaker feature Store

Fraud detection Team ✓

Credit-Risk Team ✓

Customer Retention Team ✓

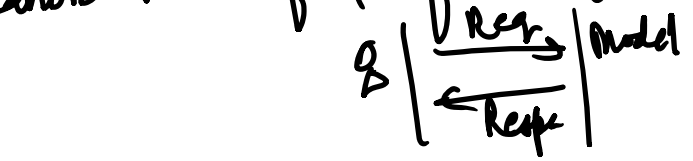
SageMaker feature "Store"
↳ create, store, discover, share, reuse features for ML.

① Feature = Measurable attribute used as input to an ML Model

② feature group = Organizes feature definitions & records.

③ offline store = Historical feature analysis & Model TRs

④ online store = Low-latency retrieval of the latest applicable feature values for (inference)



A Customer feature group

feature
- ✓ Customer-id
- ✓ TRS_count_24h
- ✓ avg_txns_value_30d

Purpose
Identify Customer records
Measure recent activity
Characterize spending behavior

feature definition = feature name, data type.

feature group = Record identifier & event-time config.

Deployment Methods

Learning your

SageMaker AI → Model Development Methods

Method	What you control	Enterprise use case
Built-in Algo	Algo config & Hyperparameters	XGBoost for customer churn
Bring your own Script	TRG logic & framework code	Custom PyTorch fraud detection model
Bring your own Container	Complete container runtime & dependencies	Specialized scientific framework

(A) Choose TRG Instance

Instance Category

1. General purpose
2. Compute optimized
3. Memory optimize
4. GPU
5. Specialized accelerators

Useful for?

Initial workloads & moderate

CPU-intensive TRGs

Large in-memory dataset

Deep learning, neural networks

Large scale TRG workloads

AutoML → SageMaker Canvas

↳ writing any code. ✗

Canvas AutoML = Create Model Button.

1. Profiles the data.
2. Generate candidate pipeline = algo based on dataset

② Generate candidate pipeline

③ Trains & evaluates

Amazon SageMaker TRG jobs : Parameter vs. Hyperparameters

① **Model parameters**
↳ Values that ML Algo learns automatically form TRG data.

② **Hyperparameters :**
↳ settings that we configure to control how the ML algo trains.

Example :

Bank → ML model → To predict, a customer will repay a loan → Yes / No.
↳ Historical Data : Income, Credit score, existing loans, Employment history, Previous loan Beha

TRG Data
Income
Credit score
→
SageMaker AI
TRG jobs
learn pattern
(Algo, para, Hyperpa)
→
TRAINED Model

Configure [Hyperparameters] = Algo learns model parameters.

What are model parameters ?

Predicted loan risk = w_1(Income) + w_2(Debt) + b

Predicted loan [illegible] = W1(

W1 = learned weight for income
W2 = learned weight for debt ✓
b = learned bias (intercept)

SDK → DS → Train (say.)
Test
valida

What are hyperparameters ?

↳ Control the TRG process / structure of model

- ✓ learning rate = steps.
- ✓ Epochs = How many times TRG passes through full dataset.
- Batch size = How many TRG examples
- # of Layers, Max TREE depth

Model Parameters

- ✓ learned from TRG Data
- ✓ Represent learned relationship.
- ✓ Biases, weights.

↓

what the model learn from Dataset

Hyperparameters

- ✓ Configure before TRG / adjust between runs
- ✓ Control the TRG Algo or Model Configuration
- ✓ learning rate, epochs, depths, 10, 3, 30

↓

How we tell the model to learn

SageMaker Training Job

↳ Managed process that runs your ML TRG code or selected algo on AWS Compute Infra.

↳ [illegible] TRG Config, Sagemaker manages

or select ... config

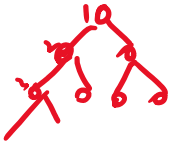
↳ Provide the TRG Config, Sagemaker manages execution environment & saves the resulting model artifacts.

TRG jobs workflow

① Prepare the dataset
↳ store TRG & validation dataset → Amazon S3

② Choose the Algo & Compute resources
XGBoost on ml.m5.xlarge instance

③ set Hyperparameter =
↳ max_depth = 5, eta = 0.1, num_round = 100



④ Run the TRG jobs

⑤ Save & evaluate the model = F1 score

	Learning rate	depth	validation F1
job 1 =	0.01	3	0.78
job 2 =	0.10	5	0.89
job 3 =	0.30	8	0.82

→ Model A V.2

Lab 8: Identify Bias in ML Models.

(1) Bias detection

(2) Model explainability

UCI Adult Income DS → XGBoost Classification Model

↓

5 Bias metrics.

more 50K — less 50K

Age, edu, occ, marital status, Gender, Race.

overall Notebook workflow

Load Adult Income DS ✓

26K X 13 cols

Pre-TRG Bias ✓

ORG data & income labels

Class Imbalance (CI)

Train XGBoost

80% = TRAIN

20% = Tests

Post_TRG Bias

↓

Model Explain